

Ginevra Cerri

SENIOR PERCEPTION ENGINEER
ROBOTICS · COMPUTER VISION · AUTONOMY

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I give robots and vehicles their sense of sight, and carry it from research prototype to shipped product. Today I run the perception workstream inside a larger autonomy stack: object detection and tracking, depth and spatial estimation recovered from cameras alone, and vision-language models that identify what the scene contains, integrated with the modules around it.

01 EXPERIENCE

Five years turning perception research into systems that run on real vehicles, with growing ownership of teams, vendors and roadmaps.

2nd
place at A2RL, the team's first eligible autonomous racing competition

250 km/h
the speed the perception stack has to stay reliable at

5+
engineers led on an external delivery, end to end

Senior Perception Engineer · SIRB AI Jan 2026 — Present · Abu Dhabi

Perception lead for an autonomy stack moving from research programme to a commercially deployed product.

- Architecting the end-to-end perception framework for the research-to-commercial transition of SIRB AI: computer-vision model optimization, deployment targets, and a large-scale data management architecture for continuous retraining.
- Running 4 concurrent sub-projects that feed the same stack, one of them with an external startup, while coordinating with the other module leads to integrate perception with the components around it.
- Applying vision-language models to classify the identity and affiliation of detected entities in the scene, alongside spatial estimation that raises object localization accuracy without depth sensors.
- Designed field demos jointly with the other sub-team leads, then validated and presented them on site, representing the perception module directly to clients.

Perception Engineer · Technology Innovation Institute Oct 2023 — Dec 2025 · Abu Dhabi

Perception lead for the TII Car Racing team, racing a driverless car wheel-to-wheel above 250 km/h.

- Led the Perception Division to a **2nd place finish in the team's first eligible A2RL competition**. Detection, tracking and long-range opponent estimation inside a hard latency budget.
- Managed a 5+ person external engineering team on a custom perception algorithm: defined technical requirements, built the dataset, integrated and benchmarked performance, and drove it through to on-track delivery.

Co-Founder & Chief Product Officer · Unchained Labs Dec 2024 — Present · Remote

- Directed end-to-end development of the Hunter ecosystem, an AI agent for on-chain insights, covering product scoping, roadmap, design direction and release, alongside a full-time engineering role.

Research Assistant, Robotics & CV · Polytechnic of Milan, mOve Sep 2022 — Jul 2023 · Milan

- Kicked off, designed and deployed the perception stack for the Aida project, which spun off into Niulinx, an autonomous-driving startup for urban mobility.
- Collaborated with Maserati S.p.A. on 1000 Miglia 2023, running the first autonomous vehicle in the race's history.

Undergraduate Researcher, CV · IZONICS Jan 2022 — Jun 2022 · Paris

- Collected and labelled 100+ samples into a training dataset for a CNN reaching 95% accuracy classifying biological vs. PVC particles; reported technical results directly to the founders.

02 SELECTED PROJECTS Four things worth asking me about: two shipped **on the road**, two shipped **on the web**.

A2RL Autonomous Racing Detection, tracking and opponent state estimation for a driverless race car. 2nd place, debut season.	2024–25	Hunter AI agent turning raw on-chain activity into readable insight. I own product and release.	2024–	1000 Miglia 2023 Deployed on a Maserati MC20 Cielo, the first autonomous vehicle ever to take part in the historic road race.	2023	Kymatics Voice-to-deployed-web-app code assistant, model agnostic. Built at Mistral Worldwide Hacks.	2024
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03 SKILLS & TOOLS The stack I build on, and how I work across teams.

PERCEPTION & ML

Computer vision Object detection & tracking Monocular depth estimation Spatial estimation & localization

Sensor fusion Vision-language models Dataset curation Model optimization & benchmarking

ENGINEERING

Python C++ ROS Docker SQL Data pipelines & MLOps MATLAB & Simulink Foxglove PlotJuggler

Git & GitHub LaTeX

LEADING & SHIPPING

Cross-team coordination External vendor management Technical requirements Product roadmapping

Client-facing demos Jira & Taiga

LANGUAGES

Italian — native English — C2, EFSET certified French — B2 Spanish — B1

04 EDUCATION

Control engineering in Milan, mechanics in Rome, a semester in Paris.

MSc Automation & Control Engineering

Sep 2020 — May 2023 · Milan, Italy

Polytechnic of Milan

Graduated 110 cum laude/110. Thesis in autonomous-vehicle perception, nominated for the Award for Best Thesis on Advanced A&C Techniques. Coursework across control systems, computer vision, machine learning and robotics.

Erasmus+ Exchange Programme

Jan 2022 — Jun 2022 · Paris, France

ESIEE Paris, Grande École d'Ingénieurs

BSc Mechanical Engineering

Sep 2016 — May 2020 · Rome, Italy

La Sapienza, University of Rome

05 RECOGNITION

Selections, awards and funding earned along the way.

Nova Talent member

2024

Curated international talent community; admitted after a multi-stage evaluation.

A2RL — 2nd place

2025

Autonomous Racing League, as perception lead of the debut campaign.

Best Thesis nomination

2023

Award for Best Thesis on Advanced A&C Techniques, Politecnico di Milano.

€16K scholarship + €1.5K prize

2020–23

Merit scholarship on admission and a graduation prize for academic excellence.

06 LEADERSHIP & BEYOND

Where I've represented others, and what fills the weekends.

Student Representative — Industrial Engineering

Jan 2017 — May 2020 · Rome, Italy

Elected voice of the mechanical engineering student body: 20+ meetings with university leadership across a three-year mandate, advocating initiatives that improved the student experience.

Outside work

Ongoing

Event photographer and private maths tutor · summer camp assistant and charity volunteer · choir member · salsa dancer · backpack traveller · modern and contemporary art enthusiast.